

A Measured Identifiability Ceiling for Rendered Scientific Images, and What It Attributes

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Abstract

Vision-language models fail on rendered scientific figures, and the failure is widely attributed to perception rather than reasoning. The evidence is usually a two-stage pipeline that replaces the model’s perception with a model-written description. That design compares two model configurations and cannot say how much information the image contains. We supply the missing quantity. A crystal-structure render is a known orthographic projection of a known structure, so the information it carries can be measured by inverting the cameras and applying a deterministic symmetry algorithm, with no model anywhere in the measurement. On a certified-label benchmark of crystal-system classification, this geometric oracle recovers 0.9524 and 0.9095 of structures on two 210-structure samples, against 0.6905 for the strongest model arm (paired per structure, discordance 61:6, $p = 1.5 \times 10^{-12}$). Every label is generated rather than annotated, so the ceiling and the ground truth come from one source. Using the oracle as one rung of a four-rung attribution ladder yields two results a two-stage comparison cannot reach. First, supplying exact geometry as text leaves 12 of the 14 scored models limited by the symbolic residual rather than by perception (median perception share 30.9%), and the share tracks model strength: “perception is the bottleneck” is false as a general statement on this task and true only of the strongest models. Second, a strong model used as the extraction stage emits syntactically perfect coordinate lists whose median recall against ground truth is 0.0000, a fabrication that downstream accuracy alone would have attributed to reasoning. The method transfers to any modality whose forward rendering is known and invertible.

1 Introduction

A model that misreads a scientific figure and a model that misreasons about it produce the same wrong answer. Separating the two requires knowing what the figure contains, and for natural images that quantity does not exist: there is no procedure that reads a photograph’s information content without a model.

Rendered scientific figures are different. A ball-and-stick crystal render is an orthographic projection of a structure whose atomic coordinates are known exactly, under a camera set the experimenter chose. The projection is invertible in a sense we make precise in Section 3: given the render’s own geometry, atom positions can be recovered by triangulating across views, and a deterministic symmetry algorithm applied to the reconstruction returns the answer the images support without consulting a model at any step. This paper builds that instrument and reports what it attributes.

The shape of the argument has prior art. Wang et al. (2025) argue that the gap on abstract-reasoning benchmarks reflects visual perception rather than inductive reasoning, and verify it with a two-stage pipeline that converts each image to a natural-language description before inducing rules. Polat et al. (2025) reach a compatible conclusion for crystallographic

images, and Yeh et al. (2025) show that multi-view geometric correspondence is a known weakness. What differs here is the exactness of the separation. In every prior design we found, the perception stage is itself a model, so the split compares two model configurations and inherits the describer’s errors, and no quantity in the design states how much of the task is answerable from the image at all. Our separation is deterministic and closed-form, and it returns a number.

Three contributions follow. (i) A certified-label benchmark with a measured identifiability ceiling. Labels are computed by a symmetry algorithm from the source structures rather than annotated, and the evaluation split excludes every chemical composition present in training, following Polat et al. (2025). The ceiling is 0.9524 rather than 1.0: the render protocol withholds a measurable amount of what the structure contains, so the gap between the ceiling and a model is separable from the gap between the ceiling and the label (Proposition 1 characterises exactly when the two coincide). (ii) A four-rung attribution ladder. Each rung removes one capability on the same named structures, all comparisons paired (Definition 2). Its outcome contradicts the perception-bottleneck reading it was built to test: with exact geometry supplied as text, the symbolic residual is the larger deficit for 12 of 14 scored models, and the perception share tracks model strength. (iii) A render-convention intervention, one leg conclusive and one inconclusive. Two of the frozen protocol’s geometric choices are measurably information-suboptimal by the oracle. A 70-structure model probe detected no corresponding change, but its discordance rates leave 7 of 16 paired comparisons unable to detect one at any outcome, so the model leg is inconclusive rather than null and is reported as such.

2 Related work

Perception versus reasoning in multimodal models. The claim that multimodal failures are perceptual rather than inferential is argued in several places. Wang et al. (2025) isolate a perception bottleneck on abstract-reasoning benchmarks via image-to-description conversion. Wu et al. (2026) decouple the two stages in post-training and find perception the profitable target; Osaka et al. (2026) characterise visual access boundaries directly; Tian et al. (2026) report that additional reasoning can reduce accuracy, as does Kancheti et al. (2026) for spatial tasks. Our ladder agrees that perception matters and disagrees about its share: with exact geometry supplied, the symbolic residual is the larger deficit for 12 of 14 scored models, and the split tracks model strength. The difference is one of measurement: every prior separation we found substitutes a model for the perception stage, so it cannot state what the image contains, whereas the oracle rung can.

Crystallography with vision–language models. Polat et al. (2025) is the closest neighbour: it stress-tests nine VLMs on rendered crystal structures under spatial- and composition-exclusion protocols. We adopt its composition-exclusion discipline and its CIF-supplied condition, citing both as prior use rather than contribution. Xu et al. (2026) benchmark VLMs on XRD peak indexing and already establish that models fail on rendered crystallographic images, which demotes our own difficulty finding to added vendor breadth on a deterministic-label target. On the model-free side, Liang et al. (2020) and Wang et al. (2024) predict symmetry from composition and from diffraction-consistent descriptors respectively; both take the opposite input to ours, so neither is comparable to our cell-metric reference.

Resolution and render protocol. Li et al. (2025) and Niu et al. (2025) show that dynamic resolution input is a live failure mode, which is why our protocol is frozen at a single resolution and why the effective-resolution audit is reported as a render-design observation rather than a finding. Multi-view correspondence is a documented weakness (Yeh et al., 2025), and view selection has been optimised end-to-end through a differentiable renderer (Hamdi et al., 2021), so we claim no novelty for optimising a render protocol. Our contribution on that axis is that two of the frozen protocol’s choices are measurably information-suboptimal by the oracle, and that an underpowered model probe detected no corresponding change.

Verifying intermediate steps. Wang et al. (2026) verify explicit visual premises to make process reward models reliable, and Qin et al. (2025) add continuous visual tokens to improve intermediate visual reasoning. Both share our premise that the intermediate must be check-

Algorithm 1 Geometric oracle $O(s, \mathcal{C})$

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1: for each view  $v$  do
2:    $Q_v \leftarrow \{(\Pi_v(Lx_i), z_i) : i = 1, \dots, N\}$   $\triangleright$  projected, species-labelled 2D points;
   cross-view identity discarded
3: end for
4: for each pair of views  $(v, w)$  and each same-species pair  $(q, q') \in Q_v \times Q_w$  do
5:    $\hat{p} \leftarrow$  intersection of the back-projected rays  $\Pi_v^{-1}(q)$  and  $\Pi_w^{-1}(q')$ 
6:   accept  $\hat{p}$  iff its projection lies within tolerance of a same-species point in every re-
   maining view
7: end for
8: return  $\text{spg}_\tau(\text{verified reconstruction})$ 

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able. The difference is that a crystal render admits an exact check against ground-truth coordinates, which is what exposes the fabrication in Section 5.

3 Setup, notation, and the geometric oracle

Structures and labels. A crystal structure is a triple $s = (L, X, Z)$: a lattice $L \in \mathbb{R}^{3 \times 3}$, fractional coordinates $X = \{x_i\}_{i=1}^N \subset [0, 1]^3$, and species $Z = \{z_i\}$. The label map is $y(s) = \text{spg}_\tau(s)$, the crystal system returned by a deterministic symmetry algorithm (spglib) at a fixed tolerance τ , taking values in the seven crystal systems. Because the label is computed rather than annotated, label correctness is not a competing explanation for model error; Section 4 describes the tolerance sweep that certifies it.

Renders. The frozen protocol renders a $2 \times 2 \times 2$ tiling of the conventional cell through five orthographic cameras $\mathcal{C} = \{c_v\}_{v=1}^5$ (three principal-axis and two oblique views) at 768 px, giving images $I_v = \mathcal{R}(s, c_v)$. The tiling is a legibility choice inherited from the render convention and is not information-free; Section 7 measures what it costs. Each camera is a known unit direction d_v with orthographic projection $\Pi_v : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $\Pi_v(p) = P_v p$ for a fixed 2×3 matrix P_v with orthonormal rows perpendicular to d_v . The task given to a model is: from $(I_1, \dots, I_5)$, name the crystal system.

The oracle. The oracle answers the same question from the renders’ own geometry instead of from pixels. It forward-projects the ground-truth atom positions of the conventional cell through each frozen camera, discards the correspondence between views, re-solves it by ray intersection with cross-view verification, and applies spg_τ to the reconstruction (Algorithm 1). It is deterministic and CPU-only. Two properties of that definition are load-bearing and we state them here rather than in the limitations. First, the oracle assumes ideal extraction: every atom’s centroid is available in every view, including atoms an image hides behind another, so R_1 bounds identifiability from the projection geometry and not from the pixels. A visibility-conditioned variant that removes occluded detections is a separate measurement and is reported in Section 7. Second, the oracle inverts the conventional-cell projections rather than the tiled ones the renderer draws; supplying it the tiled geometry instead is one of the interventions of Section 7 and costs it twenty-two structures.

Definition 1 (Identifiability ceiling). For an evaluation sample $\mathcal{S}$, the ceiling is

$$R_1 = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbf{1}[O(s, \mathcal{C}) = y(s)],$$

the fraction of structures whose label the frozen renders determine for an ideal geometric reader.

Proposition 1 (Exact recovery). Let s be a structure rendered under known orthographic cameras. Suppose (i) at least two of the camera directions are linearly independent, and (ii) the cross-view verification of Algorithm 1 admits a unique consistent correspondence, i.e. no two distinct atoms produce verified ray intersections within the matching tolerance of one another. Then the oracle recovers every atom position exactly, and $O(s, \mathcal{C}) = \text{spg}_\tau(s) = y(s)$.

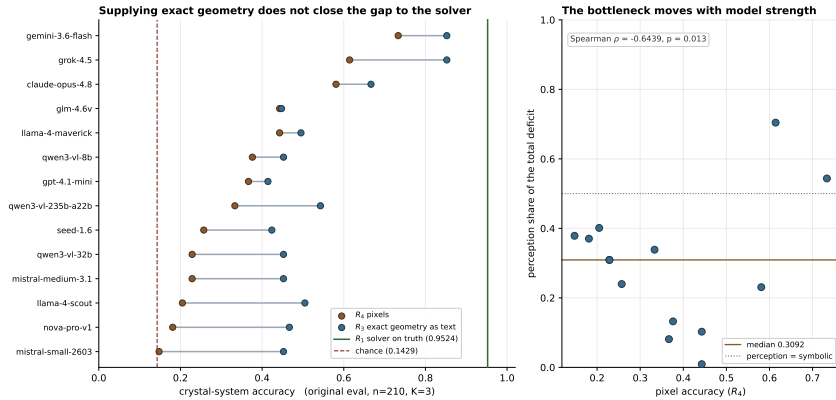


Figure 1: Left: attribution ladder, original evaluation sample ($n = 210$); rungs as in Definition 2, R_2 omitted (unscored under its pre-registered gate). Right: label ladder across four label granularities for the model-free arms; the oracle (top curve) is flat, the cell-metric baseline is not.

A proof sketch is in Appendix A; the argument is elementary (two orthographic images along independent directions determine a point as the intersection of two affine lines, and verification against the remaining views eliminates spurious matches outside the coincidence set). The proposition is useful for what its failure set says. Condition (i) holds by construction for the frozen camera set. Condition (ii) fails under projective coincidence, when distinct atoms project onto near-identical pixels, and this is exactly where the measured ceiling falls short of 1: coincidence grows with atoms per cell, which predicts the cell-size dependence observed at scale-up (Section 6). A separate quantity governs what an extractor reading pixels could recover: across the frozen five-view set, the fraction of atoms with fewer than two clear views is 0.26% (original sample) and 0.87% (replication sample), so the protocol’s occlusion leaves almost every atom triangulable in principle (Section 7).

Why a ceiling below 1.0 is the point. A design that supplied the source coordinates as text would have a ceiling of exactly 1.0 by construction and would measure nothing about the images. A ceiling below 1.0 says the protocol withholds information, and Proposition 1 locates where.

4 Benchmark construction

Certified labels. Each structure’s crystal system, Bravais lattice, point group and space group are derived from its coordinates by spg_τ at the canonical tolerance, with a tolerance sweep that quarantines any structure whose space group is not stable across it. On the kept set, label agreement with the source databases is 220/220 (one-sided exact 95% lower bound 0.9865).

Composition exclusion. No chemical composition in any evaluation set appears in training. Without this, symmetry is partly predictable from composition alone, a published task (Liang et al., 2020), and a model could score well without reading the image.

A label ladder gives a difficulty axis. Figure 1 (right) scores four labels of increasing granularity for the model-free arms. The oracle is flat across granularity because a reconstruction is either right or wrong and all four labels follow from it; the cell-metric baseline falls from 0.8952 at crystal system to 0.6810 at space group, because point group and space group depend on atom positions that lattice parameters do not encode. The identifiability claim is therefore not specific to the seven-way label.

Table 1: Attribution ladder, original evaluation sample, $n = 210$, $K=3$. The two model columns are the strongest scored model and one mid-roster model, named rather than aggregated; R_3 and R_4 within a column are the same model under two conditions, so their interval is that model’s perception component (Definition 2). R_2 is unscored: its extractor fails to triangulate on 19.0% and 22.4% of structures, over the pre-registered 5% gate. The fine-tuned arm was never run under the coordinate condition, so it carries no within-model interval and is not a column.

rung	what it removes	gemini-3.6-flash	gpt-4.1-mini
R_0 algorithm on ground-truth positions	nothing	1.0000 (definitional)	
R_1 oracle: camera inversion, then algorithm	the protocol’s losses	0.9524	
R_3 exact geometry as text ($K=3$)	the model’s perception	0.8524	0.4143
R_4 pixels ($K=3$)	nothing further	0.7333	0.3667
perception share $(R_3 - R_4)/(R_1 - R_4)$	—	0.5436	0.0813
R_2 oracle on a learned detector’s output	ideal extraction	unscored	

5 The attribution ladder

Definition 2 (Rungs and shares). For a model m evaluated on the same structures, let $R_0 = \frac{1}{n} \sum \mathbf{1}[\text{spg}_\tau(s) = y(s)]$ (the symmetry algorithm on ground-truth positions), R_1 the oracle ceiling of Definition 1, $R_3(m)$ the accuracy of m given ground-truth fractional coordinates, species and cell parameters as text, and $R_4(m)$ the accuracy of m on the pixel renders. Define the perception component $P(m) = R_3(m) - R_4(m)$, the symbolic residual $S(m) = R_1 - R_3(m)$, and the perception share $P(m)/(R_1 - R_4(m))$. R_2 denotes the oracle run on a learned detector’s output.

Three conditions make the shares interpretable, and we state them because each is checkable: the rungs are scored on the same named structures with all comparisons paired; R_3 and R_4 use the same decode budget ($K=3$ everywhere the shares are computed); and the symbolic residual bounds rather than isolates symmetry reasoning, since a residual against a deterministic solver can include failure modes other than symmetry (we test this directly below). Table 1 gives the ladder.

The top rung is definitional, and that is load-bearing. The label is spg_τ applied to the structure, so $R_0 = 1.0000$ exactly, and the $R_0 \rightarrow R_1$ interval is what camera inversion and correspondence re-resolution cost. Treating R_0 as approximately equal to R_1 would discard that interval.

Exact geometry does not close the gap. Supplying ground-truth geometry as text (the CIF-supplied condition of Polat et al. (2025), cited as prior use) lifts every model substantially, and none reaches the oracle: the best is 0.8524 against 0.9524. The pre-registered decomposition of Definition 2 gives a median perception share of 30.9% across the 14 models scored under the coordinate condition, with the symbolic residual the larger of the two for 12 of them; the two columns of Table 1 differ in perception share by a factor of six. Only the two strongest models are perception-dominated, and the share tracks pixel accuracy (Spearman $\rho = -0.6439$, $p = 0.0130$). The bottleneck moves with model strength: a more precise statement than either single attribution, and one visible only because the oracle fixes the top of the ladder.

A control pair makes R_3 readable. Removing the images and leaving only the chemical formula collapses every scored model toward seven-way chance (mean 0.1487 over the 13 models scored under the image-removal control). Removing the images and supplying full geometry lifts every scored model to between 0.4143 and 0.8524. The two conditions differ in one thing, so the lift is the geometry rather than text-mode prompting; the geometry prompts are also shorter than the five-image prompts they beat, which rules out a length artefact.

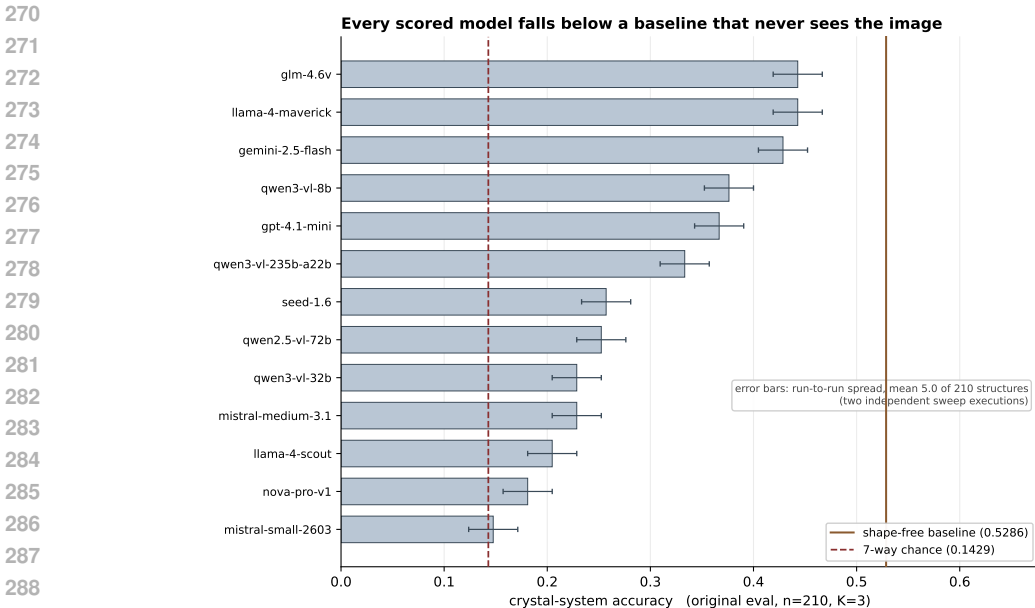


Figure 2: Zero-shot leaderboard, original sample, $n = 210$, $K = 3$, 13 scored models across eight vendors. Dashed line: the shape-free baseline (atom count, density, cell volume) on this sample. Error bars: the measured run-to-run spread of $K=3$ majority voting.

The symbolic residual is an upper bound, and we tested it. A residual against a solver may include long-list handling rather than symmetry reasoning. Regressing per-structure R_3 correctness on conventional-cell atom count over the same records (no new model calls): pooled over the 14 scored models and 2940 model-structure pairs, Spearman $\rho = -0.0908$ at $p = 8.1 \times 10^{-7}$; 13 of 14 models are negative, and accuracy falls from 0.5551 on cells at or below the median 14 atoms to 0.5129 above it (Fisher $p = 0.0241$). The pre-registered confound control does not rescue it: atom count correlates with crystal system ($\rho = -0.1866$), but the mean within-system association is -0.1053 , stronger than the pooled figure rather than attenuated, so the effect is not symmetry difficulty in disguise. We therefore report the symbolic residual as an upper bound on a symmetry-reasoning deficit, not as a measurement of one. Nothing here touches the oracle-to-model gap, which is measured against pixels.

A learned extractor fabricates its output. Substituting a strong model as the extraction stage, prompted for species and positions with the symmetry question withheld, and having a weaker model answer from that text yields 0.1429. That arm predicted two of seven classes and its accuracy equals the majority stratum’s base rate, so it measures nothing. The informative measurement is the intermediate, scored directly against ground truth: median recall 0.0000, with 105 of 206 structures having not one emitted atom within tolerance of a real one, despite a median of 48 well-formed atoms per structure. A colour-threshold blob detector achieves median recall 0.400 on the same task. Downstream accuracy alone would have read this as bad reasoning. A two-stage pipeline whose intermediate cannot be checked will attribute fabrication to reasoning, which is the argument for the instrument.

6 Where models sit

Rosters. Three conditions were run over overlapping but non-identical model sets, so we name each roster explicitly and never write a bare count. The zero-shot leaderboard scored 13 of a pre-registered 14, with moonshotai/kimi-k2.6 unscored under an endpoint-failure gate (no response on a two-structure $K=1$ probe after ten minutes, so the failure is the endpoint rather than the harness); it includes google/gemini-2.5-flash and not the newer gemini-3.6-flash that heads the ladder, which is why the leaderboard’s best row (0.4429) sits below the ladder’s best R_4 (0.7333). The image-removal control attempted 16 and scored

13, with three unscored under a pre-registered 5% API-error gate. The coordinate condition attempted 15 and scored 14, with qwen/qwen2.5-vl-72b-instruct unscored for exceeding both the error and parse gates (34.4% errors). Every unscored entry is reported unscored rather than dropped; the per-model roster table is in the supplementary.

The 13 scored zero-shot models ran on the frozen protocol with a verbatim prompt and no per-model tuning (Figure 2). The best reaches $93/210 = 0.4429$ on the original sample, far below the oracle’s 0.9524.

A baseline comparison we previously reported does not survive scale-up, and we retire it. On the original sample every model falls below a shape-free baseline reading only atom count, density and cell volume (0.4429 against 0.5286, one-sided $p = 0.0078$). We regenerated the evaluation set at $n = 1933$ under identical rules: same composition exclusion, verified zero overlap with training and both earlier samples, stratified 285 per system before quarantine, with 62 structures (3.11%) removed for label instability under the tolerance sweep. On that sample the baseline falls to 0.2054. Because the new draw also has larger cells (median 38 atoms against 14), we ran a size-matched control restricted to the original’s 95th percentile: the baseline is 0.2205 there, so the shift is not a cell-size artefact. The 0.5286 was a property of the original draw, not of the task, and the below-baseline claim is scoped to that sample rather than asserted generally.

The same control shows the oracle’s apparent drop at scale is a cell-size effect: 0.8774 on the full sample but 0.9459 at matched cell size, within 0.0065 of the original. More atoms per cell means more projective coincidence and more correspondence ambiguity, which is the failure set of Proposition 1. The ceiling-to-model gap, which is what this paper rests on, therefore survives scale-up (0.9459 against 0.4429), while the baseline comparison does not.

Run-to-run spread is a benchmark property. An unplanned second execution of the sweep gives a mean absolute difference of 5.0 structures and a maximum of 11, so adjacent leaderboard rows are not separable and the ordering is not exact.

A modality-matched reference. A cell-metric random forest reaches 0.8952 on crystal system, above every model. This reproduces an established capability rather than contributing one; it is not comparable to Liang et al. (2020), which predicts symmetry from composition alone, the opposite input and a harder problem.

The models do use the image. A benchmark on which models score poorly invites the objection that they are pattern-matching the prompt rather than reading the figure. We ran the image-absent control that Kancheti et al. (2026) made standard: the identical prompt carrying the chemical formula and no renders. Every scored model collapses toward seven-way chance (mean 0.1487), and the paired per-structure difference is significant for 10 of the 13 models scored under the image-removal control (Figure 3); the strongest model loses 0.5714. The three nulls are the three weakest models, two of which are not above chance even with images, so those are floor effects rather than evidence of formula lookup.

The strongest model arm, and what is not claimed for it. The 0.6905 figure is a supervised-then-GRPO arm on Qwen/Qwen3-VL-8B-Instruct, reported as a single-seed point estimate at $K=8$; the same adapter gives $139/210 = 0.6619$ at $K=3$. It sits inside the reference arm’s across-seed spread (0.590/0.567/0.686), exceeding that arm’s best seed by one structure at $K=8$ and trailing it by five at $K=3$, so we make no claim that this arm improves on the reference arm. It is used only as the model side of the ceiling-to-model gap (200 against 145, discordance 61:6, $p = 1.5 \times 10^{-12}$), which survives any seed in the observed spread. The three-seed extension was cut; its record and the quantitative argument for the cut are in the supplementary.

7 Render-convention interventions

Two of the frozen protocol’s geometric choices are information-suboptimal, measured by the oracle. Perturbing the cameras off the principal axes raises the ceiling from 0.9524 to 0.9952 ($p = 0.0039$, monotone: nine structures gained, none lost). Supplying the oracle the shipped protocol’s $2 \times 2 \times 2$ tiling instead of the single conventional cell costs it twenty-two

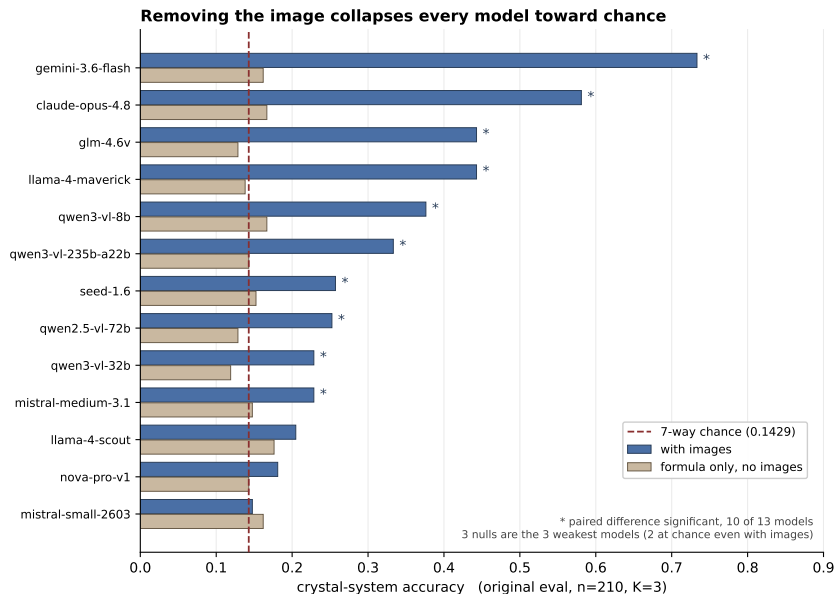


Figure 3: Image-removal control: each model answered the identical prompt with the five renders and with the chemical formula alone, paired per structure (original sample, $n = 210$, $K = 3$, 13 scored of 16 attempted). Stars mark models whose paired difference is significant; the dashed line is seven-way chance.

structures, 0.9524 against 0.8476 ($p < 10^{-6}$, gained 0, lost 22), because tiled copies project onto coincident pixels and a triangulating reader cannot separate them; both effects are instances of the coincidence failure set of Proposition 1.

Neither correction reaches the models. Four strong models across five conventions on a stratified 70-structure subsample give zero significant changes in 16 paired comparisons, with near-symmetric discordance in every convention and inconsistent direction within conventions. Power is low and we state it per comparison rather than as a single number. The test is a paired exact binomial on discordant pairs; discordance averages 6.3 of 70, and significance is reachable in only 9 of the 16 comparisons, the other 7 having too few discordant pairs for any split to reach $p < 0.05$ (Appendix B gives the procedure and the per-comparison minimum detectable differences, which run 0.0857 to 0.1286 across the 9 comparisons where significance is reachable at all, against a largest observed effect of 0.0857). We therefore do not claim the null bounds the effect at any particular size. What it supports is that no convention produced a detectable change, and that for seven comparisons the design could not have detected one. An intervention that adds information a model cannot use is evidence about the model.

What the protocol withholds is small and redundant. The oracle of Section 3 assumes ideal extraction, so a separate measurement is needed to say what the images themselves hide. Classifying every occluder by whether it hides a symmetry-equivalent copy shows most occlusion hides nothing recoverable, and because triangulation needs only two clear views, under 1% of atoms are untriangulable from the frozen view set (0.26% and 0.87%). Removing informative occlusion from the oracle’s input changes no classification at all (Figure 4), while the pre-registered redundant-only control changes several, so render adequacy is established and the deficit is forced onto the model.

8 Secondary analyses

Two axes are reported in full in the supplementary and summarised here, because both end by disclaiming themselves and neither carries a main-text claim.

Cue sufficiency. Whether the conventional-cell metric alone determines the crystal system splits the samples 140/70 and 141/69 and is a stable property of the render convention with no accuracy mechanism attached: an earlier claim that pixel accuracy simply drops on the ambiguous stratum failed four independent tests and is withdrawn. A contrast against a cell-metric control does widen on that stratum for all three frontier models on both samples (Figure 5), but the stratum is 60/70 and 58/69 hexagonal-or-trigonal, so the finding is close to a restatement of that one degeneracy, and the degeneracy-removed residual rests on $n = 10$ and $n = 12$.

One generation of model progress. It does not erase the axis (Figure 6): from gemini-2.5-flash to gemini-3.6-flash the newer model gains more on the ambiguous stratum in raw terms but closes less of its headroom to the oracle (54.1% against 64.0%), and one model pair is a comparison rather than a trend.

9 Limitations

No human baseline. Recruitment of qualified crystallographers failed. We therefore make no claim that the renders are solvable by humans; the oracle stands in for the information question only, and “adequate” means recoverable by a geometric reader.

What the oracle does not license. It reads ground-truth coordinates, not pixels. It bounds what is recoverable given ideal extraction and says nothing about achievable extraction. The extraction share of the ceiling-to-model gap is bounded from neither side: R_1 assumes perfect extraction and R_2 ’s extractor is too weak to be informative. A detector at recall above 0.9 is the most valuable missing measurement.

Sample size. All model numbers rest on two samples of 210; the minimum detectable paired difference there is near 0.09, so adjacent leaderboard rows are not separable. The model arms were not re-run at $n = 1933$ (that is 19,500 calls on a 500-structure core), so the ceiling-to-model gap at scale is bounded on the oracle side only. The oracle and both classical baselines are complete on the full 1933.

Corrections and non-recoveries. Three published values did not reproduce from their own records and are documented as non-recoveries rather than silently replaced: an earlier cue-sufficiency classifier’s exact rule, an oracle view-count curve, and the tabular classifier’s published count (twelve defensible readings of its recorded protocol span 183 to 188 without reaching the published 186). A previously published minimum-detectable-difference for the render-convention null was computed with an unpaired approximation and is retracted in favour of the per-comparison analysis of Section 7, which makes that null weaker rather than stronger. The full retraction ledger is in the supplementary (Appendix C indexes it).

10 Conclusion

When a rendering is a known projection of a known object, the information the rendering carries is computable without a model, and that number turns a benchmark into an attribution instrument. Applied here it contradicts the reading it was built to test: exact geometry leaves most models limited by the symbolic residual rather than by perception, the bottleneck moves with model strength, and a learned extraction stage can fabricate its output completely while remaining syntactically perfect. The method transfers to any modality where the forward rendering is known and invertible.

Reproducibility statement

The oracle, the label pipeline and both classical baselines are deterministic and CPU-only; the released artifact contains the render protocol, the labelling tolerances and sweep, the exact feature specification and seed for the cell-metric reference, the verbatim model prompts, and per-structure predictions for every arm reported here, so each accuracy in this paper recomputes from released vectors without API access. Every numeric claim was written by a script that reads the corresponding checkpoint record rather than transcribed, and a

verification gate refuses any manuscript sentence carrying a value absent from those records or an accuracy without its sample and decode budget. Each analysis was pre-registered before its numbers existed; the pre-registrations are released alongside the results, including for the analyses whose outcome contradicted the registered expectation. Values that did not reproduce are documented as non-recoveries (see Limitations). Dataset metadata is provided in Croissant format validated against the reference library.

Statement on the use of AI tools

This work was carried out with substantial use of an AI research assistant; this section is the disclosure, and per the author guidelines it does not count against the page limit.

Categories that apply. The assistant participated in proposing and refining hypotheses (the attribution-ladder design, the cue-sufficiency partition, the visibility conditions); designing and giving feedback on methodology and experiments (every pre-registration in the released ledger was drafted by it before the corresponding numbers existed); implementing methods (the render pipeline, the oracle, the label pipeline, every evaluation harness, every figure script); cleaning and reformatting datasets (retrieval, composition exclusion, the symmetry-tolerance quarantine); and interpreting results. Of the recommended categories it also drafted parts of this paper, created the figures, and performed literature search.

What is mechanically checked rather than asserted. Every numeric claim here is read from a checkpoint record by script; a build gate refuses any sentence carrying a value absent from those records, or an accuracy stated without its sample and decode budget. All six figures regenerate from their checkpoint records and are verified byte-for-byte. A separate linter checks the source for unescaped percent signs, unbalanced math, and undefined references. Each check was itself tested against a deliberate violation of the rule it enforces.

Errors it introduced, and how they were caught. We report these because the failure modes are the useful part. First, a published minimum-detectable-difference of 0.162 for the render-convention null was computed with an unpaired two-proportion approximation carrying no power term, for a test that was paired. Recomputing per comparison showed significance was unreachable in 7 of 16 comparisons at any outcome, which made the null weaker than the retracted figure implied, and the contribution was demoted accordingly. Second, a figure plotted parameter counts for models whose sizes are undisclosed; the axis was deleted and undisclosed-size models are now barred from any parameter axis. Third, a zero-shot sweep ran twice and an aggregation step silently mixed both runs; the leaderboard was rebuilt from single-run per-structure vectors and the second run is published as a labelled replication whose run-to-run spread is reported as a property of the benchmark. Further withdrawals, including three documented non-recoveries, are verbatim in the supplementary’s retraction record.

What a human is accountable for. The authors take responsibility for every claim in this paper, including those produced or checked by the assistant, and for the decision about which results ship and which are reported as withdrawn.

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A Proof sketch for Proposition 1

Let atom a have Cartesian position $p \in \mathbb{R}^3$ and let v, w be two views with linearly independent directions d_v, d_w . Its images are $q = P_v p$ and $q' = P_w p$. The back-projections are the affine lines $\ell_v = \{x : P_v x = q\}$ (direction d_v) and $\ell_w = \{x : P_w x = q'\}$ (direction d_w). Both contain p ; since $d_v \nparallel d_w$, two non-parallel lines in $\mathbb{R}^3$ that intersect do so in exactly one

point, so $\ell_v \cap \ell_w = \{p\}$ and the intersection step of Algorithm 1 recovers p exactly from the correct pairing. It remains to show no incorrect pairing survives. A spurious pairing (q, q'') with q'' the image of a different atom b either yields no intersection (skew lines, rejected) or yields a point $\hat{p} \neq p$ whose projections must, by hypothesis (ii), fall outside the matching tolerance of any same-species point in at least one remaining view, so the verification step rejects it. The verified reconstruction is therefore exactly the atom set, on which spg_τ returns $y(s)$ by the definition of the label. Condition (ii) is where the argument is not vacuous: it fails precisely on projective coincidence, when distinct atoms project within tolerance of one another in the verifying views. The measured ceiling gap (0.0476 on the original sample) and its growth with atoms per cell (Section 6) are measurements of that failure set.

B Statistical procedures

Paired comparisons. Every model-versus-model and model-versus-oracle comparison on a shared sample is a paired exact binomial (sign) test on discordant structures: with g structures one arm alone gets right and l the other alone, the two-sided p is that of $\text{Bin}(g+l, 1/2)$ evaluated at $\min(g, l)$. For the render-convention probe, the per-comparison minimum detectable difference is the smallest $|g-l|/n$ reaching $p < 0.05$ given that comparison’s observed $g+l$; with $g+l \leq 5$ no split reaches it, which is the source of the 7-of-16 unreachable comparisons in Section 7. Label certification. The one-sided exact 95% lower bound for k/k agreements is $0.05^{1/k}$; for $k = 220$ this is 0.9865. Run-to-run spread. The second sweep execution is retained as a labelled replication; spread is reported as the mean and maximum absolute per-model difference in correct counts.

C Index of corrections and non-recoveries

The full retraction ledger, verbatim and in checkpoint order, is in the supplementary. Main-text items: the retired below-baseline claim (Section 6), the retracted unpaired minimum-detectable-difference (Section 7), the withdrawn stratified-accuracy mechanism (Section 8), and the three documented non-recoveries (Limitations).

D Supporting figures

The visibility-conditioned oracle of Section 7 and the two secondary axes of Section 8 are stated numerically in the main text; their figures are placed here.

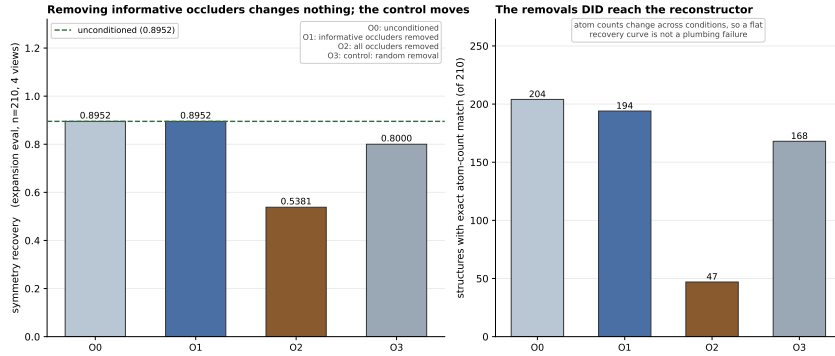


Figure 4: Visibility-conditioned oracle: recovery when informative occlusion is removed from the oracle’s input, when redundant occlusion only is removed (pre-registered control), and unconditioned, on both evaluation samples.

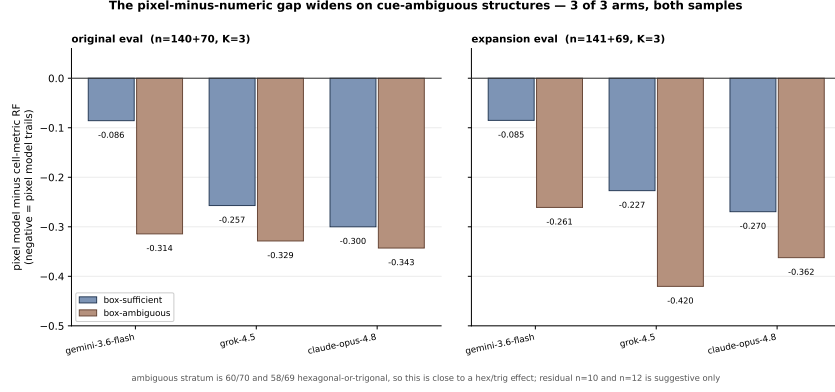


Figure 5: Pixel accuracy minus a cell-metric random forest on the same structures, by cue-sufficiency stratum, three frontier models, both evaluation samples ($K = 3$). Negative means the pixel model trails the numeric reader. Confound recorded on the figure: the ambiguous stratum is 60/70 and 58/69 hexagonal-or-trigonal.

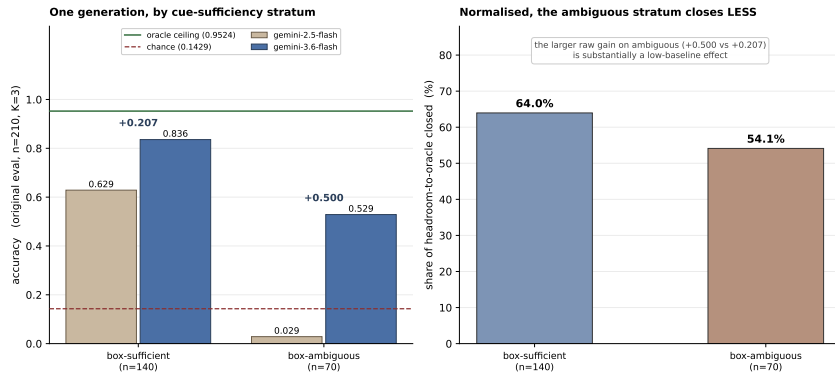


Figure 6: One generation of model progress, original sample, $n = 210$, $K = 3$. Left: raw accuracy by cue-sufficiency stratum, with the oracle ceiling and seven-way chance drawn. Right: the same gains normalised by headroom to the oracle.